Name: Dereck Lu
GitHub Username: lu1336
Purdue Username: lu1336
Instructor: Sturm
Section:

Problem 1

1.

Given the probability density function:

$$f_X(x) = \begin{cases} \frac{3}{4}(1-x^2) \; when \; x \in [-1,1] \\ 0 \qquad\quad otherwise \end{cases}$$

If you happen to plot this distribution, then by looking at the plot, determine by inspection, what is the mean of the distribution.

Mean of the distribution:

0

From this distribution we grab 100 i.i.d samples $X_1, X_2, \ldots X_{100}$.

The sample mean random variable is defined as

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

What is the shape of the probability density function of the sample mean?

Normal

What is the mean of the probability density function of the sample mean?

0

Explain how you know the shape and mean.

The original distribution is symmetric about 0 so its mean is 0. By the Central Limit Theorem, the sample mean of 100 i.i.d. samples is approximately normal. Its mean equals the mean of the original distribution, which is 0.

Part B

Let X be a normally distributed random variable with mean = 0 and variance = 1. What is the distribution of the random variable Z = 3X+1 (Give your answer with the shape, mean and variance).

What is the shape of the distribution?

Normal

What is the mean?

1

What is the variance?

9

Problem 2

You are given data in *city_vehicle_survey.txt* representing the average age of vehicles across various counties. The transportation department claims the average age of vehicles is 5 years. You are tasked with testing this claim.

1. Formulate null and alternative hypotheses for a statistical test that seeks to challenge this belief. What are the null and alternative hypotheses?

   1. Null Hypothesis:

   H0: mu = 5 (the average age of vehicles is 5 years)

   2. Alternative Hypothesis:

   H1: mu != 5 (the average age of vehicles is not 5 years)

   3. What type of test should be used and why?

   A two-sided one-sample z-test since we are checking if the mean is different from 5 and the sample size (1024) is bigger than 29.

2. Carry out this statistical test using the *city_vehicle_survey.txt* sample. Report the sample size, the sample mean, the standard error, the standard score (z or t, depending on what was used), and the p-value.

   ********ROUND ALL DECIMAL VALUES TO 4 DECIMAL PLACES******

Sample size	1024
Sample mean	5.1141
Standard error	0.0614
Standard score	1.8585
p – value (if less than 0.01 use scientific notation)	0.0631

   Are the results statistically significant at a level of 0.05?

   Yes **(No)**

   What (if anything) can we conclude about the hypothesis at the confidence level of 0.05?

   The p-value (0.0631) is bigger than 0.05 so we fail to reject H0. We do not have enough evidence to say the average vehicle age is different from 5 years, but this does not prove that it is 5.

Are the results statistically significant at a level of 0.10?

Yes **No**

What (if anything) can we conclude about the hypothesis at the confidence level of 0.10?

The p-value (0.0631) is smaller than 0.10 so we reject H0. At this level there is evidence that the average age of vehicles is different from 5 years.

3. What is the largest standard error for which the test will be significant at a level of 0.05? What is the corresponding minimum sample size? (You may assume that the population variance and mean does not change.)

********ROUND ALL DECIMAL VALUES TO 4 DECIMAL PLACES******

Largest standard error	0.0582
Corresponding minimum sample size	1139

4. Suppose the transportation department believes the mean vehicle age is the same in counties with and without emission control programs. Two datasets, *vehicle_data_1.txt* (with emission programs) and *vehicle_data_2.txt* (without emission programs), are used to test this assumption.

   1. Null Hypothesis:

   H0: mu1 = mu2 (the mean vehicle age is the same with and without emission programs)

   2. Alternative Hypothesis:

   H1: mu1 != mu2 (the mean vehicle ages are different)

   3. What type of test should be used and why?

   A two-sample two-sided z-test, because there are two independent samples and both sample sizes are large (512 and 868).

5. Carry out this statistical test using the *vehicle_data_1.txt* population and *vehicle_data_2.txt* population samples. Report the sample sizes, the sample means, the standard error, the z-score, and the p-value. Are the results significant at levels 0.05 or 0.10? What (if anything) can we conclude about the hypothesis at the two different confidence levels?

********ROUND ALL DECIMAL VALUES TO 4 DECIMAL PLACES**** ******

Sample size of *vehicle_data_1* (Emission)	512
Sample size of *vehicle_data_2* (Without Emission)	868
Sample mean of *vehicle_data_1* (Emission)	5.5415
Sample mean of *vehicle_data_2* (Without Emission)	6.2482
Standard error	0.1044
Standard score	-6.7721
p – value (if less than 0.01 use scientific notation)	1.2695 x 10^-11

1. Are the results statistically significant at a level of 0.05?

Yes **No**

2. Are the results statistically significant at a level of 0.10?

Yes **No**

3. What (if anything) can we conclude (i.e., what is the interpretation of the result)?

The p-value is about 1.2695 x 10^-11, which is much smaller than both 0.05 and 0.10, so we reject H0 at both levels. There is very strong evidence that the mean vehicle age is different between counties with emission control programs and those without. The mean age with emission programs (5.5415) is lower than without (6.2482), so counties with emission programs tend to have newer vehicles on average.

Problem 3

1. Use the sample to construct a 90% confidence interval for the average sodium of snacks. Report whether you will use a z-test or t-test and report the sample mean, the standard error, the standard statistic (t or z value), and the interval. (Think, which distribution should you use here if very few data points are available?)

A t-test (t distribution with 19 degrees of freedom), because the sample is small (n = 20) and the population standard deviation is unknown.

********ROUND ALL DECIMAL VALUES TO 4 DECIMAL PLACES******

Sample mean	140.4500
Standard error	1.3426
Standard score (t or z value)	1.7291
90% confidence interval	(138.1285, 142.7715)

2. Repeat Q1 for a 95% confidence interval.

********ROUND ALL DECIMAL VALUES TO 4 DECIMAL PLACES******

Standard error	1.3426
Standard score (t or z value)	2.0930
95% confidence interval	(137.6400, 143.2600)

Is your interval wider or narrower compared to using the 90% confidence interval in Q1?

Wider **Narrower**

3. Repeat Q2 if you are told that the population standard deviation is 5.
Will you use a t-test or z-test (Hint: Think which distribution should you use here now that you have the true population standard deviation)? Justify your answer.

A z-test, because the population standard deviation is known (5), so we use the standard normal distribution.

********ROUND ALL DECIMAL VALUES TO 4 DECIMAL PLACES******

Standard error	1.1180
Standard score (t or z value)	1.9600
95% confidence interval	(138.2587, 142.6413)

Is your interval wider or narrower than the interval computed in Q2?

Wider **Narrower**